ERP – Project Execution & Resource Plan

Agentic Multimodal eKYC Verification, Risk Fusion and Escalation Agent (HSR-AGAI-029)

Advanced AI / Agentic AI course | Prepared 10 November 2026 | Team: 1–2 students

> **How to read "ERP" here.** The request asked for an "ERP doc for doing the project". I have treated it as an **Execution & Resource Plan**: who does what, with which data, tools and compute, by when, with what risks and acceptance gates. Items taken from the project bank or study material are marked **[Bank]** or **[Study]**. Items that are my planning suggestions are marked **[Plan]** and should be confirmed with your faculty.

1. Project Charter

Item	Detail
Project ID and title	HSR-AGAI-029 – Agentic Multimodal eKYC Verification, Risk Fusion and Escalation Agent **[Bank]**
Industry reference	FRS Labs (identity verification and fraud detection for banks); not an official sponsor **[Bank]**
Area	Identity, Fraud Detection, Cybersecurity **[Bank]**
Team and duration	1–2 students; one semester, about 10–14 weeks of implementation **[Bank]**
Plan used below	12 weeks **[Plan]**
Core agent loop	Observe → reason/plan → act through a tool/API → observe outcome → improve/escalate **[Bank]**
Extension chosen	Adds an **audio branch** (spoken challenge + voice anti-spoof) and text consistency checks, so the input is Vision + Text + Audio **[Plan]**

Goal statement. Build a deployable agent that fuses face, document, liveness, text and voice evidence, decides *approve / retry / manual-review / reject*, requests more verification when needed, and sends uncertain or high-risk cases to a human verifier. **[Bank, extended]**

2. Scope

In scope

- Multimodal perception: face matching, OCR of a synthetic ID document, liveness/deepfake check, spoken-challenge ASR, replay-artefact voice check, form-text consistency.
- Semantic state and risk fusion with calibrated risk score.
- LLM/SLM reasoning with RAG over a KYC policy/runbook corpus and a case-history memory.
- ReAct controller with multi-hop investigation and a stopping policy.
- RL-style decision policy over approve / retry / manual-review / reject.
- HITL reviewer console, ground-truth capture, feedback loop and fine-tuning pipeline.
- API, dashboard, tests, containerized cloud deployment, basic monitoring.

Out of scope

- Real customer data or company-private data (the bank assumes public data or a documented synthetic dataset) **[Bank]**.
- Any claim of regulatory certification. Policy documents are synthetic or paraphrased and must be labelled so.
- Production-grade scale, 24×7 operations, and legal sign-off.

3. Deliverables and Acceptance Criteria

The deliverable list comes from the bank's "Minimum B.Tech Deliverables" for this project **[Bank]**; owners, weeks and acceptance tests are **[Plan]**.

#	Deliverable	Owner	Due	Acceptance test
D1	Git repository with source code	Both	Continuous; final W12	Fresh clone runs with one documented command
D2	Documented dataset and preprocessing	A	W2	Data card lists source, licence, size, labels, splits
D3	Baseline and improved method	A	W5	Both evaluated on the same held-out split
D4	Quantitative evaluation: FAR/FRR, ROC-AUC, document extraction F1	A	W5, refreshed W11	Metrics reproduced from a script and logged in MLflow

#	Deliverable	Owner	Due	Acceptance test
D5	Agent architecture and state-action diagram	B	W6	States, actions, tools, guardrails all labelled
D6	API/backend service	B	W9	Endpoint tests pass; p95 latency measured
D7	Dashboard / reviewer console	B	W10	Reviewer can approve, modify or reject a queued case
D8	Tool-call / action trace	B	W7 onward	Every case has a stored, replayable trace
D9	Guardrail and HITL test cases	Both	W9	All escalation triggers fire in tests
D10	Agent benchmark and failure-analysis report	Both	W11	Scenario suite run; failures categorized
D11	README, architecture diagram, deployment instructions	B	W11	Another student can deploy from the README
D12	Final report	Both	W12	Follows the companion report outline
D13	Demo video and presentation	Both	W12	Shows one approve, one retry, one reject-with-HITL case

4. Work Breakdown and 12-Week Timeline **[Plan]**

Week	Focus	Key outputs
1	Requirements and architecture freeze	Job definition, state/action list, repo, issue board
2	Data preparation	Four Kaggle datasets downloaded; 200 synthetic identity bundles built; splits fixed (140/30/30 by identity)
3	Vision perception baselines	Face matching, OCR extraction, liveness/deepfake scorer

Week	Focus	Key outputs
4	Audio and text branches	Spoken-challenge ASR, simple replay-artefact voice score, name/DOB fuzzy matching
5	Semantic state and risk fusion	Fusion model, calibration, baseline vs improved metrics
6	RAG and memory	Policy corpus indexed, case-history store, SLM/LLM selected
7	ReAct controller	Tools wrapped as APIs, multi-hop chains, stopping policy
8	Decision policy	Simulator, Q-learning policy, rule guardrails
9	Backend, HITL and persistence	FastAPI services, PostgreSQL schema, reviewer queue, ground-truth logging
10	Dashboard and feedback loop	React console, feedback table, fine-tuning pipeline, MLflow registry
11	Deployment, benchmark, failure analysis	Docker images, CI/CD, cloud deployment, monitoring, benchmark run
12	Documentation and demo	Final report, video, presentation, buffer

Milestone gates

- **G1 (end W2):** data card approved and architecture frozen.
- **G2 (end W5):** baseline and improved perception metrics available.
- **G3 (end W8):** closed-loop agent runs end to end on the simulator.
- **G4 (end W10):** HITL console and feedback loop working.
- **G5 (end W11):** deployed, benchmarked, failure analysis written.

5. Resource Plan

5.1 People and roles **[Plan]**

Role	Student A (Perception and ML)	Student B (Agent and Platform)
Primary	Data, vision/audio models, fusion, evaluation, fine-tuning	RAG, ReAct, RL policy, API, UI, deployment
Secondary	Test data for guardrails	Metrics logging, MLflow

Role	Student A (Perception and ML)	Student B (Agent and Platform)
Joint	Architecture, benchmark design, report, demo	

If you are a solo team: keep the full loop but cut breadth. Use one document type, drop the voice anti-spoof model (keep the spoken-digit ASR check), use a rules-plus-bandit decision policy instead of full RL, and deploy on a single container host. Confirm this reduction with faculty, which the project bank explicitly allows **[Bank, Faculty Note]**.

5.2 Data resources

The project uses **one simple synthetic set of 200 identity bundles**, built from four free Kaggle datasets. Each bundle holds a selfie, an ID card image, form text and a spoken four-digit challenge, plus a known correct decision. This keeps the whole project laptop-sized. **[Plan]**

Component	Kaggle source	Use
Face pairs	LFW	Use people with two or more photos: one photo goes on the ID card, another is the selfie. A different person's photo gives an impostor pair
Spoken digits	Free Spoken Digits	Join four digit clips into the challenge audio; wrong digits or added noise make spoof audio
ID card images	Recaptured identity documents and the turabbajeer profile (synthetic passports and licences; the exact dataset is to be picked from the profile)	Card layouts and recapture realism. Text on each card is generated by the team, so OCR ground truth is known
Liveness and spoof	Anti-spoofing live (sample)	Live and attack examples; print and screen artefacts are simulated for the rest

Bundle type	Count	How it is built	Expected decision
Genuine, clean (Applicant A)	120	Two real photos of one person, clean card, correct digits	Approve

Bundle type	Count	How it is built	Expected decision
Genuine, poor capture (B)	30	Blur and glare added to the card, borderline face match	Retry, then approve
Spoof (C)	30	Print or screen artefacts on the selfie, replay-style audio, wrong digits, one device ID shared across several bundles	Reject through HITL
Mismatch	20	Another person's selfie, or form text that differs from the card	Manual review or extra verification
Total	**200**	Split by identity: 140 train, 30 validation, 30 test	

A simulated reviewer, driven by the known labels with some noise added, supplies the HITL verdicts.

Data rules: Kaggle needs a free login. Check each page's licence and size before downloading, and confirm that the vendor's sample allows augmentation. LFW photos show real people: use them only for face pairs, put generated text on every card, label each bundle synthetic, and do not republish. No real ID numbers. Keep raw, processed and label folders separate and versioned. Sizes on the Kaggle pages were not verified for this plan, but none should approach MIDV-2020 (124 GB) or IDNet (about 400 GB), which are left out. Voice anti-spoofing is reduced to a replay-artefact check plus the spoken-digit challenge, and device/session metadata is generated.

The feedback loop is demonstrated with simulated reviewer verdicts on the 140 training identities, released in small batches, because the 500-label retraining trigger in the report is larger than this dataset.

5.3 Software and tools

Stack from the bank: Python, FastAPI, PostgreSQL, Docker, Git/GitHub, React/Next.js, MLflow, cloud deployment, an agent-orchestration framework or equivalent, a vector store for RAG, and tool/API adapters **[Bank]**.

Layer	Tool choice **[Plan]**
Vision / audio / OCR models	Open-source pretrained models, fine-tuned where needed
SLM/LLM	One small open model for local runs; optional hosted API for comparison

Layer	Tool choice **[Plan]**
RAG	Vector store (for example pgvector or FAISS) over the policy corpus
Agent orchestration	A framework or a hand-written ReAct loop; keep the loop inspectable
Backend	FastAPI, PostgreSQL
Frontend	React/Next.js reviewer console
MLOps	MLflow for experiments and model registry; GitHub Actions for CI
Packaging	Docker and docker-compose
Monitoring	Basic request/latency/error metrics and a dashboard panel

5.4 Compute

Need	Suggested provision **[Plan]**
Model training and fine-tuning	One GPU session (campus lab or free/low-cost notebook GPU)
Inference during demo	CPU-capable path for the SLM and OCR; GPU optional
Hosting	One small cloud VM or container service
Storage	Small object store or disk volume for media; PostgreSQL for metadata

5.5 Effort and cost

Item	Estimate **[Plan]**
Effort	About 24 person-weeks for two students over 12 weeks (less per head if solo, with the scope cuts above)
Software licences	Rs. 0 target; use open-source tools

Item	Estimate **[Plan]**
Compute and hosting	Team to estimate from the chosen provider's current price list; target free tiers or institutional credits
Contingency	Hold about 1 week of the timeline as buffer (W12)

6. Governance and Ways of Working **[Plan]**

- **Cadence:** weekly planning, a mid-week sync, and a faculty review at each gate.
- **Git workflow:** feature branches, pull-request review by the other student (or self-review checklist if solo), protected main branch, CI running unit tests.
- **Definition of done:** code, test, documentation and a logged metric or trace for every task.
- **Decision log:** one file recording each architecture decision, the alternatives considered and the reason.
- **Experiment discipline:** every training run logged in MLflow with data version, parameters and metrics.

7. Risk Register **[Plan]**

#	Risk	Likelihood	Impact	Mitigation
R1	Dataset licence or privacy problem	Medium	High	Use only public or synthetic data; check each Kaggle page's licence (vendor samples may bar derivative use) and record it in the data card
R2	Too few spoof/deepfake samples	High	Medium	Generate replay/print variants; report limits honestly in the failure analysis
R3	LLM/SLM hallucination in policy answers	Medium	High	Answer only from retrieved policy text; groundedness check; send low-groundedness cases to HITL

#	Risk	Likelihood	Impact	Mitigation
R4	Agent loops or over-investigates	Medium	Medium	max_steps, confidence threshold, no-new-information rule, loop detection **[Study]**
R5	Bias across demographic groups in face matching	Medium	High	Report FAR/FRR per group where labels exist; do not claim fairness beyond what is measured
R6	Tool/API failure during a case	Medium	Medium	Timeouts, retries, safe fallback to MANUAL_REVIEW
R7	Compute shortage	Medium	Medium	Smaller models, shorter clips, cache embeddings
R8	No real reviewers for HITL	High	Medium	Simulated reviewer plus faculty/peer reviewers; log that labels are simulated
R9	Scope creep from the audio branch	Medium	Medium	Treat voice anti-spoof as stretch; spoken-digit ASR is the minimum
R10	Deployment fails late	Medium	High	Containerize from W3; deploy a hello-world service by W6

8. Quality and Test Plan

Test layer	What is tested	Pass condition
Unit	Each perception function and tool adapter	Expected output on fixed fixtures
Model	FAR/FRR, ROC-AUC, document F1, ASR word error rate	Meets or beats the baseline on the held-out split
Agent	Benchmark scenarios: clean pass, blurry retry, deepfake, mismatched name, tool failure, conflicting evidence	Correct decision, correct tool sequence, grounded rationale
Guardrail/HITL	Every escalation trigger	Case lands in the reviewer queue with its trace

Test layer	What is tested	Pass condition
Integration	Upload to decision to reviewer to feedback	Full loop completes and is logged
Non-functional	End-to-end latency and cost per case	Within the budget set in G3

9. Deployment and Handover Checklist

- [ ] Services containerized (API, agent, model server, database, console)
- [ ] Configuration and secrets kept outside the repository
- [ ] Database migrations scripted
- [ ] CI builds images and runs tests
- [ ] Deployed to cloud with a public or faculty-accessible URL
- [ ] Health check and basic metrics visible
- [ ] Rollback path documented (previous image tag and previous model version)
- [ ] README lets a new person deploy in under an hour
- [ ] Demo data and demo script included

10. Success Metrics **[Bank]**

Family	Metrics
Model	FAR/FRR, ROC-AUC, document extraction F1
Agent	Task/action success, tool success/error rate, groundedness or decision-trace quality
Human oversight	Human override / escalation rate
Efficiency	End-to-end latency and cost

Targets are to be set by the team at G2 from baseline results; none are promised in this plan.

11. Assumptions and Open Points

1. "ERP" is read as Execution & Resource Plan. If your course expects a different template, the sections above can be remapped.

2. The audio branch is an extension beyond the bank's original description of this project (face matching, OCR, liveness/deepfake checks, risk fusion).
3. The data plan is one 200-identity synthetic set built from four Kaggle datasets (section 5.2); confirm each page's licence and size. Hosting and compute choices are suggestions.
4. Schedule assumes 12 weeks within the bank's 10–14 week range.

Sources: Advanced AI / Agentic AI Project Bank (HSR-AGAI-029 brief and course requirements); Combined AI Study Materials (top-down agentic design, stopping policy, benchmarking).